

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	SWTID-2026-4457
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User & Role Management	Create a test user (EEE User) in ServiceNow Create custom roles bb1, bb2, bb3 and bb4 Assign all roles to the EEE User
FR-2	Custom Table Creation	Create the table Institution Details (u_institution_details) Add fields: Student Roll Number, Student Name, Faculty Name, Branch (ECE, EEE, CSE), Email, Phone Number, Description Create sample records with different branch values
FR-3	Read ACL (Script-Controlled)	Create a record-level Read ACL with Advanced = true and role bb1 Add data condition Branch is EEE Add script that returns true for admin or bb1, otherwise false Show only EEE records to bb1 users; show all records to admin; show no records to users without the role
FR-4	Create ACL	Create a Create ACL requiring role bb2 Display the New button only to users with the required roles
FR-5	Write ACL	Create a Write ACL requiring role bb3 Allow users with bb1, bb2 and bb3 to edit EEE branch records
FR-6	Delete ACL	Create a Delete ACL requiring role bb4 Allow users with bb1 to bb4 to delete EEE branch records
FR-7	Access Verification	Impersonate the EEE User and the Admin to open the Institution Details list Confirm access is granted only when the ACL script evaluates to true

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Roles, ACLs and records are managed through the standard ServiceNow interface using simple steps (Users, Roles, Tables, Access Control). Users without access see a clear "No records to display" message.
NFR-2	Security	Record-level security is enforced by script-controlled Read, Create, Write and Delete ACLs. Access is granted only when the ACL script evaluates to true (role + Branch = EEE); all other users are denied. ACL changes require the security_admin role.
NFR-3	Reliability	The ACL script gives the same result for the same user and role every time, with a default deny (return false) for any user who does not meet the conditions.
NFR-4	Performance	ACL evaluation uses simple role checks (gs.hasRole) and a data condition, so lists of the Institution Details table load without noticeable delay.
NFR-5	Availability	The solution runs on the ServiceNow instance, so the ACLs are available whenever the instance is available.
NFR-6	Scalability	More roles, branches (e.g. ECE, CSE), records and users can be added by creating new roles and ACLs or updating the script and data condition, without redesigning the table.